

Jialong Wu

(+86) 13587027920 | wujialong0229@gmail.com | manchery.github.io | [Google Scholar](https://scholar.google.com/citations?user=...)

EDUCATION

Ph.D. in Software Engineering

School of Software, Tsinghua University

Aug 2022 – June 2027 (*expected*)

Beijing, China

- Advisor: [Prof. Mingsheng Long](#)
- Research Interests: World Models, Reinforcement Learning

Bachelor in Software Engineering

School of Software, Tsinghua University

Aug 2018 – June 2022

Beijing, China

- GPA: 3.98/4.0, Rank: 1/89

Bachelor in Pure and Applied Mathematics (*Second Degree*)

Department of Mathematical Sciences, Tsinghua University

Aug 2019 – June 2022

Beijing, China

- GPA: 3.75/4.0

EXPERIENCE

Research Intern

ByteDance Seed

March 2025 – *present*

Beijing, China

PUBLICATIONS AND PREPRINTS

Seed1.8 Model Card: Towards Generalized Real-World Agency

Bytedance Seed: ... , Jialong Wu (*Contributor*), ... *Technical report*, 2026.

[\[Project website\]](#) [\[PDF\]](#)

Visual Generation Unlocks Human-Like Reasoning through Multimodal World Models

Jialong Wu, Xiaoying Zhang#, Hongyi Yuan, Xiangcheng Zhang, Tianhao Huang, Changjing He, Chaoyi Deng, Renrui Zhang, Youbin Wu, Mingsheng Long#. *arXiv preprint*, 2026.

[\[Project website\]](#) [\[PDF\]](#) [\[Code\]](#) [\[Slides\]](#) [\[Datasets\]](#)

Thoth: Mid-Training Bridges LLMs to Time Series Understanding

Jiafeng Lin*, Yuxuan Wang*, Jialong Wu*, Huakun Luo, Zhongyi Pei#, Jianmin Wang. *arXiv preprint*, 2026.

[\[PDF\]](#) [\[Code\]](#) [\[Models\]](#)

JEPA-VLA: Video Predictive Embedding is Needed for VLA Models

Shangchen Miao, Ningya Feng, Jialong Wu, Ye Lin, Xu He, Dong Li, Mingsheng Long#. *arXiv preprint*, 2026.

[\[PDF\]](#)

Audio-Visual World Models: Towards Multisensory Imagination in Sight and Sound

Jiahua Wang*, Shannan Yan*, Leqi Zheng*, Jialong Wu, Yaoxin Mao. *arXiv preprint*, 2025.

[\[PDF\]](#)

Vid2World: Crafting Video Diffusion Models to Interactive World Models

Siqiao Huang*, Jialong Wu*, Qixing Zhou, Shangchen Miao, Mingsheng Long#. *ICLR*, 2026.

[\[Project website\]](#) [\[PDF\]](#) [\[Code\]](#)

Foresight Diffusion: Improving Sampling Consistency in Predictive Diffusion Models

Yu Zhang*, Xingzhuo Guo*, Haoran Xu, Jialong Wu, Mingsheng Long#. *ICLR*, 2026.

[\[PDF\]](#)

RLVR-World: Training World Models with Reinforcement Learning

Jialong Wu, Shaofeng Yin, Ningya Feng, Mingsheng Long#. *NeurIPS*, 2025.

[\[Project website\]](#) [\[PDF\]](#) [\[Code\]](#) [\[Models & Datasets\]](#) [\[Poster\]](#)

Seed1.5-VL Technical Report

Dong Guo, Faming Wu, Feida Zhu, Fuxing Leng, Guang Shi, ... , Jialong Wu (*Contributor*), ... *Technical report*, 2025.

[\[Project website\]](#) [\[PDF\]](#) [\[Code\]](#)

Trajectory World Models for Heterogeneous Environments

Shaofeng Yin*, **Jialong Wu***, Siqiao Huang, Xingjian Su, Xu He, Jianye Hao, Mingsheng Long#. *ICML*, 2025.
[PDF] [Code]

CompilerDream: Learning a Compiler World Model for General Code Optimization

Chaoyi Deng*, **Jialong Wu***, Ningya Feng, Jianmin Wang, Mingsheng Long#. *KDD*, 2025.
[PDF] [Code]

Long-Sequence Recommendation Models Need Decoupled Embeddings

Ningya Feng*, Junwei Pan*, **Jialong Wu***, Baixu Chen, Ximei Wang,
Qian Li, Xian Hu, Jie Jiang, Mingsheng Long#. *ICLR*, 2025.
[PDF] [Code]

iVideoGPT: Interactive VideoGPTs are Scalable World Models

Jialong Wu*, Shaofeng Yin*, Ningya Feng, Xu He, Dong Li, Jianye Hao, Mingsheng Long#. *NeurIPS*, 2024.
[Project Website] [PDF] [Code] [Models] [Poster] [Slides] [Blog]

HarmonyDream: Task Harmonization Inside World Models

Haoyu Ma*, **Jialong Wu***, Ningya Feng, Chenjun Xiao, Dong Li, Jianye Hao, Jianmin Wang, Mingsheng Long#. *ICML*, 2024.
[PDF] [Code] [Poster] [Slides] [Talk] [Blog]

Pre-training Contextualized World Models with In-the-wild Videos for Reinforcement Learning

Jialong Wu*, Haoyu Ma*, Chaoyi Deng, Mingsheng Long#. *NeurIPS*, 2023.
[PDF] [Code] [Poster] [Slides] [Blog]

CLIPood: Generalizing CLIP to Out-of-Distributions

Yang Shu*, Xingzhuo Guo*, **Jialong Wu**, Ximei Wang, Jianmin Wang, Mingsheng Long#. *ICML*, 2023.
[PDF] [Code]

Supported Policy Optimization for Offline Reinforcement Learning

Jialong Wu, Haixu Wu, Zihan Qiu, Jianmin Wang, Mingsheng Long#. *NeurIPS*, 2022.
[PDF] [Code] [Poster] [Slides]

Out-of-Dynamics Imitation Learning from Multimodal Demonstrations

Yiwen Qiu, **Jialong Wu**, Zhangjie Cao, Mingsheng Long#. *CoRL*, 2022.
[PDF] [Code]

Flowformer: Linearizing Transformers with Conservation Flows

Haixu Wu, **Jialong Wu**, Jiehui Xu, Jianmin Wang, Mingsheng Long#. *ICML*, 2022.
[PDF] [Code]

HONORS AND AWARDS

ByteDance Scholarship (awarded to 20 students in China and Singapore) BytaDance [Certificate]	2025
ICLR 2025 Notable Reviewer ICLR 2025 [Link]	2025
National Scholarship (Top 2%) Ministry of Education of the People's Republic of China [Certificate 2020] [Certificate 2024]	2020, 2024
Excellent Teaching Assistant Award Tsinghua University [Certificate]	2024
NeurIPS 2023 Top Reviewer NeurIPS 2023 [Link] [Certificate]	2023
Future Researcher Scholarship Tsinghua University [Certificate]	2022
Outstanding Graduates (Top 2%) School of Software [Certificate] & Tsinghua University [Certificate] & Beijing [Certificate]	2022

Excellent Graduation Thesis Tsinghua University [Certificate]	2022
Comprehensive Excellence Award (Top 10%) Tsinghua University [Certificate 2019] [Certificate 2021] [Certificate 2023]	2019, 2021, 2023, 2025
34th National Olympiad in Informatics (NOI 2017), Silver Medal (81st Place) China Computer Federation (CCF) [Certificate] [Link]	July 2017

INVITED TALKS AND ACTIVITIES

World Models and Human-like Reasoning MERL Seminar Series	March, 2026
World Models and Human-like Reasoning Google DeepMind London	Feb, 2026
A Path Towards General World Models BAAI Conference , Beijing, China	June, 2025
Panelist AIFC Side Event of the AI Action Summit 2025 , Paris, France	Feb, 2025
Towards General World Models: Pre-training, Multi-Modality, and Scalable Architecture ByteDance Research	June, 2024
Intrinsic Task Harmonization Inside World Models RLChina	Feb, 2024

PROFESSIONAL SERVICES

Reviewer | PC Member

- International Conference on Learning Representations (*ICLR*) 2024 – 2026
- Conference on Neural Information Processing Systems (*NeurIPS*) 2023 – 2025
- International Conference on Machine Learning (*ICML*) 2022 – 2026
- Conference on Computer Vision and Pattern Recognition (*CVPR*) 2026
- Transactions on Machine Learning Research (*TMLR*)

Teaching Assistant

- Deep Learning, Fall 2023, Tsinghua University. Instructor: [Prof. Mingsheng Long](#).
- Introduction to Algorithms, Spring 2023, Tsinghua University. Instructor: [Prof. Bin Wang](#).
- Machine Learning, Fall 2022, Tsinghua University. Instructor: [Prof. Mingsheng Long](#).
- Introduction to Artificial Intelligence, Spring 2021, 2022, 2024, Tsinghua University. Instructor: [Prof. Mingsheng Long](#).

Student Counselor

- Undergraduate Student Counselor, [Dushi College](#), Tsinghua University, 2024–2027 (*expected*).

SELECTED PROJECTS

MiniVeo3-Reasoner: Thinking with Videos from Open-Source Priors 170+ Stars We reproduce chain-of-frames visual reasoning by fine-tuning open-source video models.	2025
Awesome Multi-task Learning 760+ Stars A curated list of datasets, codebases, and papers on Multi-Task Learning (MTL).	Up-to-date
ImageLabeler C++, Qt 120+ Stars, 40+ Forks Image annotation tool, supporting 2D/3D detection/segmentation annotation.	Summer 2019
Incremental Decision Tree Python (NumPy, Matplotlib, Pandas) Re-implementation of VFDT and EFDT, reproducing paper <i>Extremely Fast Decision Tree</i> .	Fall 2020

TECHNICAL SKILLS

Programming Language: Python, C/C++, MATLAB, Java, HTML/CSS, JavaScript

Professional Software: PyTorch, JAX, NumPy, Git, LaTeX, Vue

Language: China (native), English

Last updated: March 27, 2026